ECON2050

Revision Lecture

University of Queensland



Final Exam

The information provided here is intended for the final exam

Final Exam - Logistics

- Date: 16th November 2026
- Time: See personalised exam timetable
- Venue: See personalised exam timetable
- **Make sure you arrive early to your exam venue on the day!**
- Bring your student ID
- Bring a UQ approved calculator

Final Exam - Format

- Working time: 120 mins (2hrs)
- Planning time: 10 mins
- Answer booklet: Gradescope answer booklet
- **Make sure you write in the designated spaces provided.**
- **Answers written outside the designated areas will not be marked**
- Students should not disassemble the answer booklet. This may result in answers corresponding to the wrong question. It is the student's responsibility to answer in the appropriate spaces for each question (or part of).
- If you use the additional pages at the back, please indicate to the markers that you did. Eg: For Q3(a), please see additional pages at the back

Final Exam Information

- **Refer to the course profile**
- Weighting: 52%
- Coverage: The entire course
- The exam cover sheet is posted on blackboard
- Closed book exam
- One A4 sheet of handwritten or typed notes, double sided, is permitted

Format

- There are 5 short answer questions with sub-questions
- Total marks: 70 marks
- Consists of theory, calculations, interpretation, graphs...etc

Study Tips/General Tips

- Be mathematically rigorous as possible. This will provide clarity of what you mean.
- Eg: "*The prices are positive*", "*The firm should/must produce something*"...etc may not be clear, despite my best effort here.
- Let p denote prices and q denote quantities
- $\{(p, q) \in \mathbb{R}^2 : p > 0, q \geq 0\}$
- Here, it shows that prices are strictly greater than 0, and hence positive.
- q is non-negative, greater than or equal to 0.
- $q \geq 0$ avoids the ambiguity of "should/must" *produce something*

Study Tips/General Tips

- The lectures provide clearer notation than tutorials, so use this as a guide.
- Assignments are also a guide for notation, interpretations, question style, format...etc. This is a good place to start
- Tutorials provide the main ideas and examples
- Past Exams papers?

Past Exams

- Past exams can be used as practice/study
- However, due to changes in recent semesters, some of the previous content may not be applicable this semester. So use at your own discretion.
- I do not have solutions to past exams.

Review

- Limits, continuity, graphing/sketching (curves, set, functions...)...etc have been recurring themes.
- These may be indirectly assessed. Eg: *"we need the function f to be continuously differentiable"* to apply a certain theorem.
- Should be able to apply definitions, check and show if it is satisfied or not
- Multivariate calculus is another major theme. Differentiation will be required and assumed to be well-understood.
- Basic economics is assumed, in particular what we have discussed, marginal product of capital/labour, utility functions, production functions, shadow prices...etc
- Black-Scholes, arbitrage theory, options pricing theory...etc are not required
- These were just lecture examples of linear systems
- If you are required specific context in a particular subject area, it will be provided in the question

Review

- Convexity, concavity and surrounding materials, such as line segments and some earlier parts of the course (eg: interior points, boundary, exterior points...etc) should be well understood
- This is because they are recurring throughout the course
- Optimisation (both un/constrained) is another major theme. This will be required and assumed to be well-understood.
- These will be mostly practical questions with applications.
- Already, you might have noticed the spirit of the assignments coincide with what's in this (non-exhaustive) list

Thank you for the semester and
Good luck!

Q1 a) midpoint: half way.

$$[(2,-1),(-4,5)]$$

$$\frac{1}{2}(2,-1)+\frac{1}{2}(-4,5)=(-1,2)$$

For $(0,1)$ we need.

$$\lambda(2,-1)+(1-\lambda)(-4,5)$$

$$\Rightarrow \left(2\lambda+(1-\lambda)(-4),\ -\lambda+(1-\lambda)(5)\right)$$

$$\Rightarrow (6\lambda-4,\ 5-6\lambda)=(0,1)$$

$$6\lambda-4=0$$
$$\lambda=4/6=2/3$$

$$5-6\lambda=1$$
$$\Rightarrow \lambda=2/3$$

Since $2/3 \in [0,1]$, the point $(0,1)$ is on the line segment.

Q1

c) $\lambda = \frac{1}{2}$ $[(-3,2),(5,6)]$ | $(9,8)$

$$\left(\frac{-3+5}{2}, \frac{2+6}{2}\right) = (1,4)$$

$$\lambda(-3,2) + (1-\lambda)(5,6)$$

$$(5-8\lambda, 6-4\lambda)$$

$5 - 8\lambda = 9$

$\Rightarrow \lambda = -1/2$

~~$6 - 4\lambda = 8$~~

$6 - 4(-\frac{1}{2}) = 8$

$(9,8)$ both coordinates agree.
The point $(9,8)$ is on the line, but $\lambda \notin [0,1]$, so it's not on the segment. It lies beyond the end point $(5,6)$

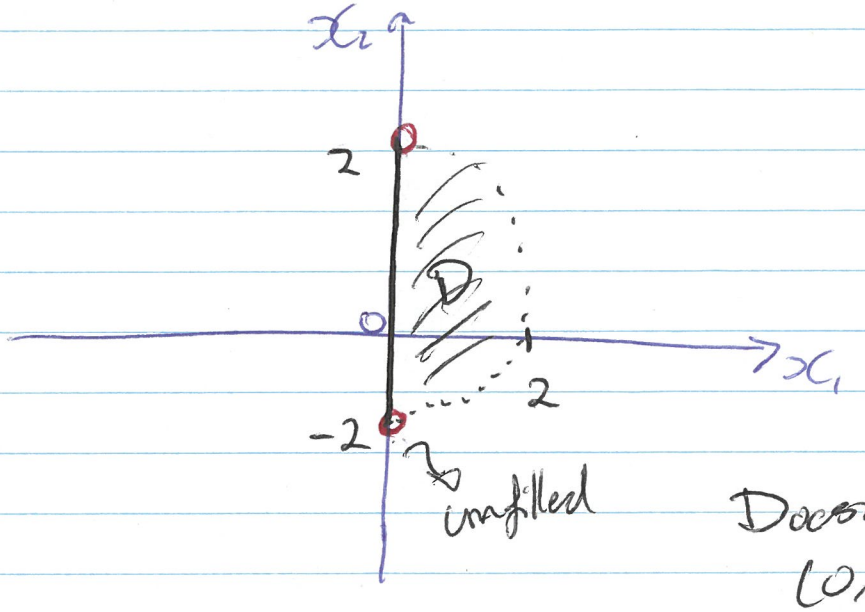
Q2
a) $f(x_1, x_2) = \ln(4 - x_1^2 - x_2^2) + \sqrt{x_1}$

$4 - x_1^2 - x_2^2 > 0$ & $x_1 \geq 0$

$\therefore D = \{(x_1, x_2) \in \mathbb{R}^2 : x_1^2 + x_2^2 < 4, x_1 \geq 0\}$

↳ circle centred @ origin radius 2

★ Note: the notation



Doesn't include (0,2) & (0,-2)

Check (0,2)

$0^2 + 2^2 = 4 \not< 4$ (not strictly less than 4)

(0,-2)

$0^2 + (-2)^2 = 4 \not< 4$ " "

Q2

b) $f(x_1, x_2) = \frac{1}{x_1 - x_2} + \sqrt{x_1 x_2}$

★ $\frac{1}{x_1 - x_2}$ requires $x_1 - x_2 \neq 0$

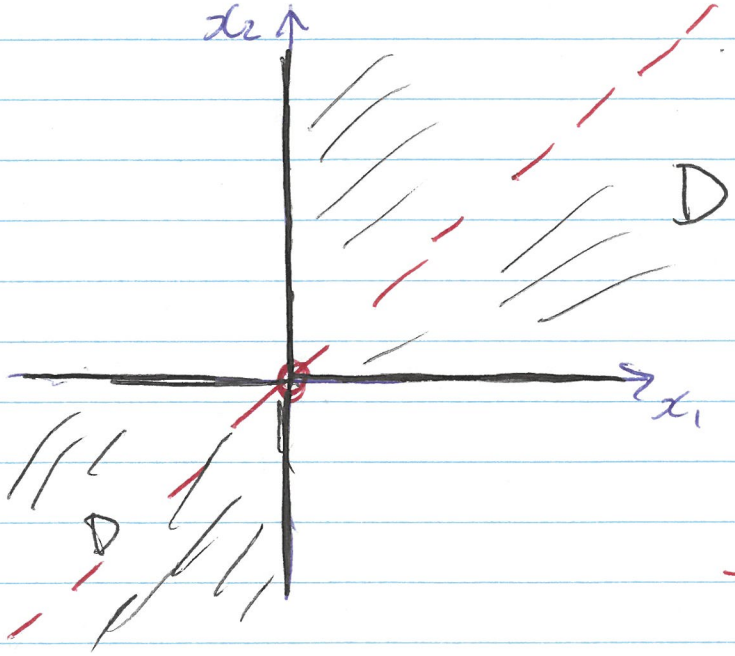
$\therefore x_1 \neq x_2$

★ $\sqrt{x_1 x_2}$ requires $x_1 x_2 \geq 0$

↳ ① x_1 & x_2 have the same sign

② when atleast one of them is 0

$$D = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 x_2 \geq 0,\ x_1 \neq x_2\}$$



(0,0) excluded

Q3

$$f(x,y) = x^3y^2 - 4xy + \ln(y)$$

$$\frac{\partial f(x,y)}{\partial x} f_x = 3x^2y^2 - 4y$$

$$\frac{\partial f(x,y)}{\partial y} = 2yx^3 - 4x + \frac{1}{y}$$

$$\frac{\partial f(x,y)}{\partial x \partial y} = 6yx^2 - 4$$

$$\frac{\partial f(x,y)}{\partial y \partial x} = 6yx^2 - 4$$

$$\frac{\partial f(x,y)}{\partial y^2} = 2x^3 - \frac{1}{y^2}$$

$$Hf(x,y) = \begin{pmatrix} 6xy^2 & 6yx^2 - 4 \\ 6yx^2 - 4 & 2x^3 - \frac{1}{y^2} \end{pmatrix}$$

Cross-partial derivatives are the same, so Young's theorem holds

$$Hf(1,1) = \begin{pmatrix} 6 & 2 \\ 2 & 1 \end{pmatrix}$$

Q4 $f(x,y) = -2x^2 + 2xy - y^2 + 4x$

$f_x = -4x + 2y + 4 = 0 \Rightarrow -2x = 4 \Rightarrow x = -2$

$f_y = 2x - 2y = 0 \Rightarrow x = y$

$\therefore (2,2)$ is a critical point

$f(2,2) = -2(2^2) + 2(2)(2) - (2^2) + 4(2)$

$-8 \quad +8 \quad -4 \quad +8 = 4$

Note: $\mathbb{R}^2$ is open & convex

f is concave

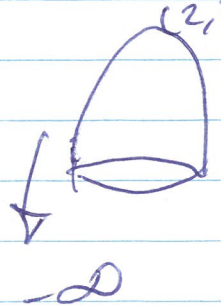
$(2,2)$ is the global (& local) maximum

Every local extremum in the interior must be a critical point. So no other local extremum.

There is no minimum.

Fix $x = 0$

$f(0,y) = -y^2 \to -\infty$



Q5

(a) (i)

Recall: A set S of vectors is linearly independent if

$$\sum_{j=1}^{n} a_j \vec{x}^j = \vec{0} \text{ implies } a_j = 0 \ \forall j = 1, \dots, n$$

& not linearly indep if

$$\exists\, a_1, \dots, a_n \in \mathbb{R}, \text{ not all } 0$$

$$\text{s.t. } \sum_{j=1}^{n} a_j \vec{x}^j = \vec{0}$$

$$\{(1,2), (3,6)\}$$

$$3(1,2) - (3,6) = \vec{0}$$

$\therefore$ linearly dependent

Rank = 1